Integrated Barangay Complaint Classification and Management Workflow

Live complaint processing and controlled offline model maintenance

A. Live Complaint Processing

Resident Submits Complaint
Subject and narrative; supporting context if available

Meaning-Preserving NLP
Prepare privacy-conscious narrative for text features

Fitted Word + Character TF-IDF
Transform text using saved vectorizers

Saved, Calibrated Linear SVM
Predict Valid / Suspicious / Spam

Predicted Suspicious or Spam?

No: Valid

Yes

Accepted Case Management
Unresolved / Ongoing / Resolved

Accepted after review

Human-in-the-Loop Review
Verify or correct TEXT label
Accept / Clarify / Reject independently

Clarify / reject

Non-Accepted Outcome Record
Pending clarification or documented rejection

B. Separate Operational and Training Records

Case and Action Records
Disposition; assignments; deadlines; actions; referrals; verified resolution

Versioned Human-Verified Labels
Narrative version; verified class; correction reason; audit/provenance

C. Controlled Offline Model Maintenance

Quality Check and Consolidate Feedback
Only eligible verified narrative labels

Retraining justified and authorized?

No

No Candidate Training Run
Keep collecting verified feedback and retain deployed model

Yes

Train, Calibrate and Validate Candidate
Consolidated training data; fixed validation protocol

Passes release criteria?

No

Retain Deployed Model
Archive failed candidate; current version remains

Yes

Approve and Release Versioned Pipeline
NLP + TF-IDF + SVM + calibrated model

Approved model for future predictions only

Solid arrows: case flow Dashed arrows: verified training feedback and approved model release